# Anahita Bolourani

**Research Scientist**

Mechanistic Interpretability | Activation Steering | Safety & Trustworthiness Evaluation | Causal Inference | Efficiency

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## Summary

Researcher in mechanistic interpretability, representation steering, and LLM safety, with a Ph.D. from UCLA and a background in statistics and machine learning. Uses linear probes and activation patching to localize emotion-related representations across transformer layers, and causal interventions to steer model behavior (ACL 2025). Co-developed PsySET (ACL 2026), a benchmark evaluating the effectiveness and unintended safety consequences of steering LLM emotions and personality traits.

## Research Experience

**Researcher** 2024 – Present

USC Institute for Creative Technologies (ICT)

- **Mechanistic Interpretability (ACL 2025, Findings).** Built linear probes and activation-patching experiments that localized emotion inference to middle-layer attention and MLP sublayers across 10 instruction-tuned LLMs from five families (1B–14B), with activation ablations providing complementary causal evidence. In Llama 3.2 1B, steering hidden states along orthogonalized, probe-derived appraisal directions shifted generated emotions as appraisal theory predicts: raising other-agency promoted anger, lowering it promoted guilt. *Cited in Anthropic's "Emotion Concepts and their Function in a Large Language Model" (2026).*
- **Steering and safety (ACL 2026, oral).** Psyset: a benchmark for steering LLM emotions and personality: compared prompting, vector injection, and LoRA fine-tuning (SFT, DPO) across eight emotions and the Big Five on four models (4B–70B). Evaluated unintended effects on six TrustLLM dimensions (safety, truthfulness, fairness, robustness, privacy, ethics) and showed that joy steering can raise jailbreak success and reduce refusals in Llama 3.1 8B Instruct, while anger steering can raise toxicity.
- **Alignment and reasoning (COLM 2025).** Showed on about 26,000 synthetic social events that most tested LLMs form impressions mainly from the actor's identity, whereas Affect Control Theory also weighs the behavior and the recipient, and that post-training (SFT, DPO, RLVR) shifts those judgments in unintended ways.
- **Reasoning and steerability (ICLR 2026 workshop).** Compared thinking and non-thinking LLMs on rational-choice axioms and emotion-conditioned decisions, finding that stronger preference coherence can coexist with greater susceptibility to affective steering.
- **Evidence-adaptive stopping for LLM agents (ARR October 2026, under review).** Trained a Qwen3-8B agent to decide, turn by turn, whether its evidence suffices to commit to a forecast or to keep observing. Distilled a 122B teacher's evidence-grounded trajectories, then trained it with PPO under explicit observation and reasoning-token costs plus a probability-calibration advantage. The learned policies reached **0.65 commit-point AUPRC versus 0.31–0.33 for GPT-5 and GPT-6 Sol**, with lower calibration error (ECE 0.08–0.16 vs. 0.22–0.25) and ∼60% fewer reasoning tokens than frozen open-weight baselines.

**Researcher** 2025 – Present

TrustworthyAI Lab, UCLA

- **TableRouter (ARR October 2026, under review).** Built a training-free framework that decides, question by question, what a frozen LLM should see from a long table: statistics computed over the full table, or a budgeted set of rows and column summaries. Across three LLMs and three long-table benchmarks, it beat the strongest of six baselines in all nine settings by 1.2–18.3 points, using 77–95% fewer estimated FLOPs than truncated-table prompting. With the answer-determining rows held fixed, adding computed statistics doubled Qwen2.5-7B accuracy (40.0% to 80.0%). Moreover, on a post-freeze evaluation whose questions were committed before any model call, TableRouter scored 35.8% against 16.4% for the strongest baseline.
- **RL fine-tuning of a diffusion model (graduate coursework).** Fine-tuned TabSyn, a pre-trained latent-diffusion tabular generator, for downstream classifier utility with KL-regularized REINFORCE and PPO-style clipped-surrogate updates (adaptive KL trust region, baseline variance reduction); switching the reward from negative entropy to classifier confidence raised synthetic-test AUC from 0.78 to 0.85.

**Graduate Research Assistant** 2022 – 2025

B. John Garrick Institute for the Risk Sciences, UCLA

NSF-Funded Caltech–UCLA Collaboration

- Built an end-to-end Python monitoring and signal-processing platform for continuous triaxial streams from Community Seismic Network sensors at NASA JPL, automating ingestion, quality control, event selection, FFT/PSD frequency extraction, dashboards, and recurring reports for the project team.

- Led a statistical evaluation of U.S. vertical seismic-load provisions over 19,316 California sites and 3,000+ ground-motion scenarios; the findings supported evidence-based recommendations for national design criteria.

**Associate Instructor** 2023 – 2026
UCLA Samueli School of Engineering

- Taught probability, statistics, optimization, and technology economics, law, and ethics across ten quarters

## Selected Publications

** Equal contribution.*

- Banayeeanzade, A., Tak, A. N., Bahrani, F., **Bolourani, A.**, Blas, L., Ferrara, E., Gratch, J., & Karimireddy, S. P. "Psychological Steering in LLMs: An Evaluation of Effectiveness and Trustworthiness," *Proceedings of ACL 2026 (Long Papers)*, oral.
- Tak, A. N., Banayeeanzade, A., **Bolourani, A.**, Kian, M., Jia, R., & Gratch, J. "Mechanistic Interpretability of Emotion Inference in Large Language Models," *Findings of ACL 2025*.
- Tak, A. N.*, **Bolourani, A.***, Shank, D. B., & Gratch, J. "Impact of LLM Alignment on Impression Formation in Social Interactions," *Conference on Language Modeling (COLM 2025)*.
- Tak, A. N.*, Banayeeanzade, A.*, **Bolourani, A.***, Bahrani, F., Chaubey, A., Karimireddy, S. P., Schwarz, N., & Gratch, J. "Sparks of Rationality: Do Reasoning LLMs Align with Human Judgment and Choice?" *ICLR 2026 Workshop on Representational Alignment*.
- **Bolourani, A.**, Lin, X., Kwok, T. S. T., & Cheng, G. "TableRouter: Query-Aware Evidence Construction for Long-Table Question Answering," *under review, ACL Rolling Review (October 2026)*.
- **Bolourani, A.**, Tak, A. N., & Gratch, J. "Knowing When to Step In: LLMs Must Learn How To Reason About Human Disputes," *under review, ACL Rolling Review (October 2026)*.

## Education

**University of California, Los Angeles (UCLA)** 2025 – Present
M.S., Statistics and Data Science, 2025 – Present

**University of California, Los Angeles (UCLA)** June 2026
Ph.D., Engineering, June 2026

## Technical Skills

**Interpretability & Steering:** linear probing, activation patching, causal interventions, activation steering, LLM safety and trustworthiness evaluation, controlled behavioral experiments.

**Machine Learning:** PyTorch, Transformers, vLLM; LLM post-training evaluation, LoRA, teacher distillation, DPO, PPO, KL-regularized policy gradients, GRPO, diffusion models, retrieval and indexing (RAG)

**Experimental Rigor:** causal inference, experimental design, paired bootstrap confidence intervals, Holm correction, calibration analysis (ECE, Brier), held-out and pre-committed evaluation, data-leakage audits

**Programming & Tools:** Python, SQL, R; Git, Docker, Jupyter, Google Colab; LLM coding agents.

## Selected Honors

Jane Permaul Women in Leadership Scholarship (2024) | ASCE Los Angeles Outstanding Graduate Student Award (2023, 2024) | ASCE Le Val Lund Award (2024)